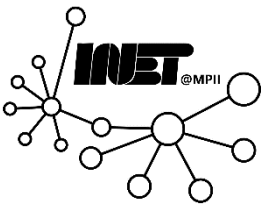




Routing: BGP

Devashish Gosain



Inter-AS routing: BGP



The *de facto* standard: **Border Gateway Protocol (BGP)**

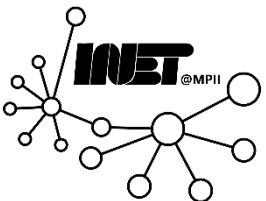
BGP provides each **AS** a means to:

- Obtain subnet reachability information from neighboring ASs
- Propagate reachability information to all routers in the AS
- Determine “*good*” routes to subnets based on reachability information and routing policy.

Allows a subnet to advertise its existence to rest of the Internet: “*I am here*”

Issues:

- Which routing algorithm?
- How are routes advertised?
- How to implement routing policies?



BGP-4



- BGP = Border Gateway Protocol
- Is an exterior routing protocol (EGP)
- Is a Policy-Based routing protocol
- Is the de facto EGP of today's global Internet
- Has a reputation for being complex
- Supports hierarchical routing
- Is a distance vector protocol



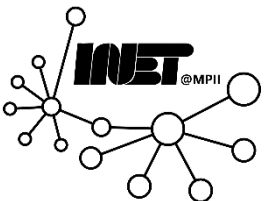
BGP history



- 1989: BGP-1 [RFC 1105]
 - Replacement for EGP (1984, RFC 904)
- 1990: BGP-2 [RFC 1163]
- 1991: BGP-3 [RFC 1267]
- 1995: BGP-4 [RFC 1771] (only 57 pages!)
 - Support for CIDR

Changes primarily driven by scalability issues

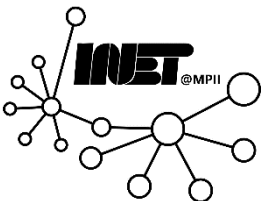
Development dominated by Cisco



Routing tasks: BGP



- Neighbor?
 - Discovery
 - Maintenance
- Database?
 - Granularity
 - Maintenance – updates
 - Synchronization
- Routing table?
 - Metric
 - Calculation
 - Update



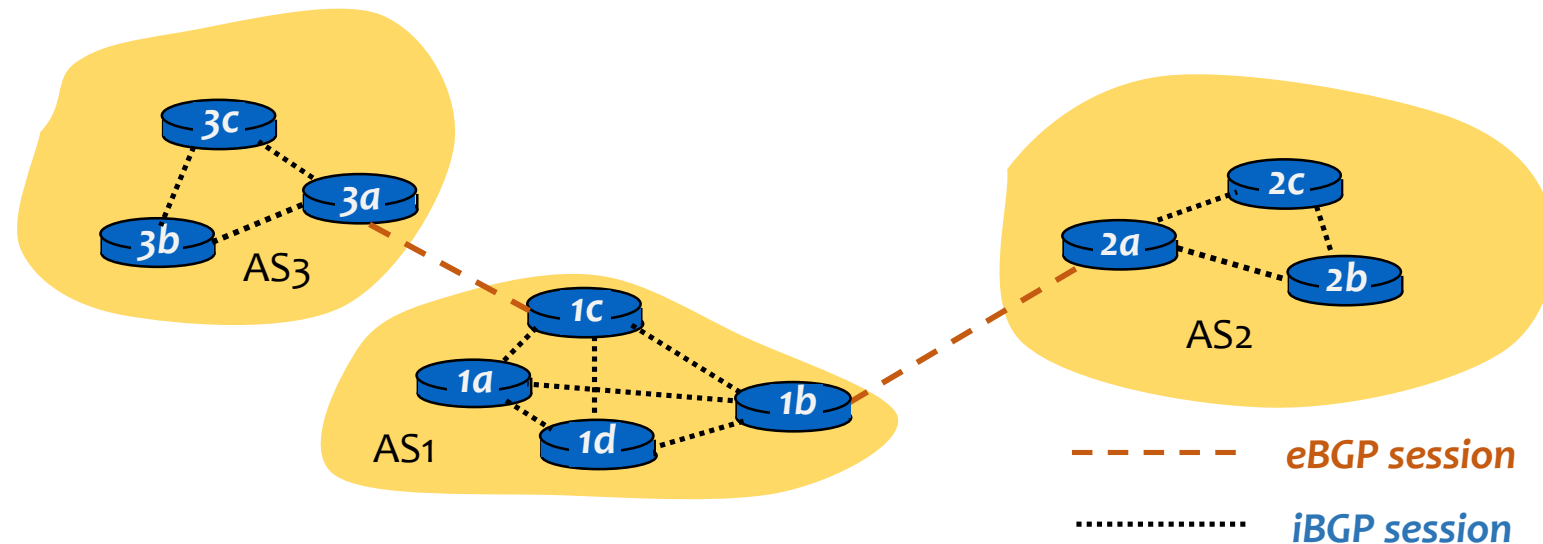
BGP Basics



Pairs of routers (**BGP peers**) exchange routing info over **semi-permanent TCP** connections:

BGP sessions

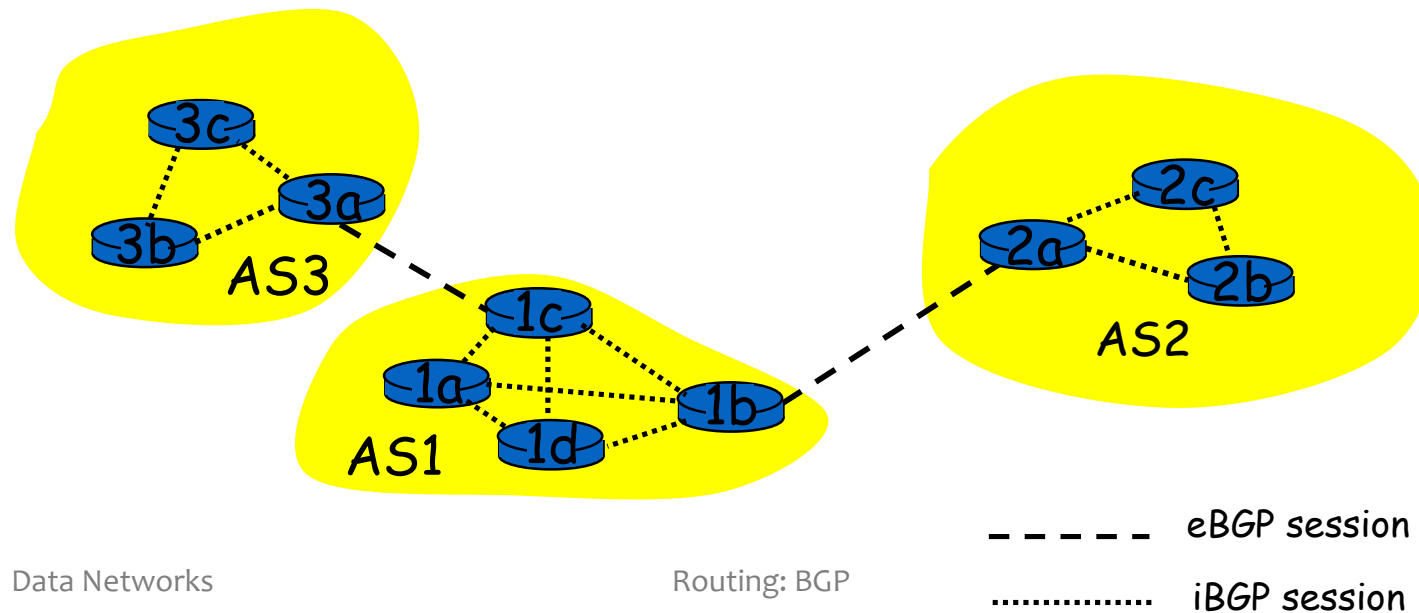
- Note that BGP sessions do not correspond to physical links.
- When **AS2** advertises a prefix to **AS1**, **AS2** is promising it will forward any datagrams destined to that prefix towards the prefix.
 - **AS2** can aggregate prefixes in its advertisement



Distributing reachability info



- With eBGP session between 3a and 1c, AS3 sends prefix reachability Info to AS1
- 1c then use iBGP to distribute this new prefix reach. Info to all routers in AS1
- 1b then re-advertise the new reach. Info to AS2 over 1b-to-2a eBGP session
- When routers learn about a new prefix, they create entries for the prefix their forwarding table

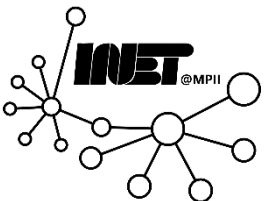


Routing policy



Reflects goals of network provider

- *Which routes to accept from other ASes?*
- *How to manipulate the accepted routes?*
- *How to propagate routes through network?*
- *How to manipulate routes before they leave the AS?*
- *Which routes to send to another AS?*



Policies with BGP



BGP provides capabilities for enforcing various policies

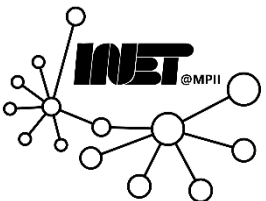
- Policies are **not** part of BGP!
- Policies are *used to configure* BGP
- BGP *enforces* policies by ...
 - Choosing paths from multiple alternatives
 - Controlling advertisements to other ASes



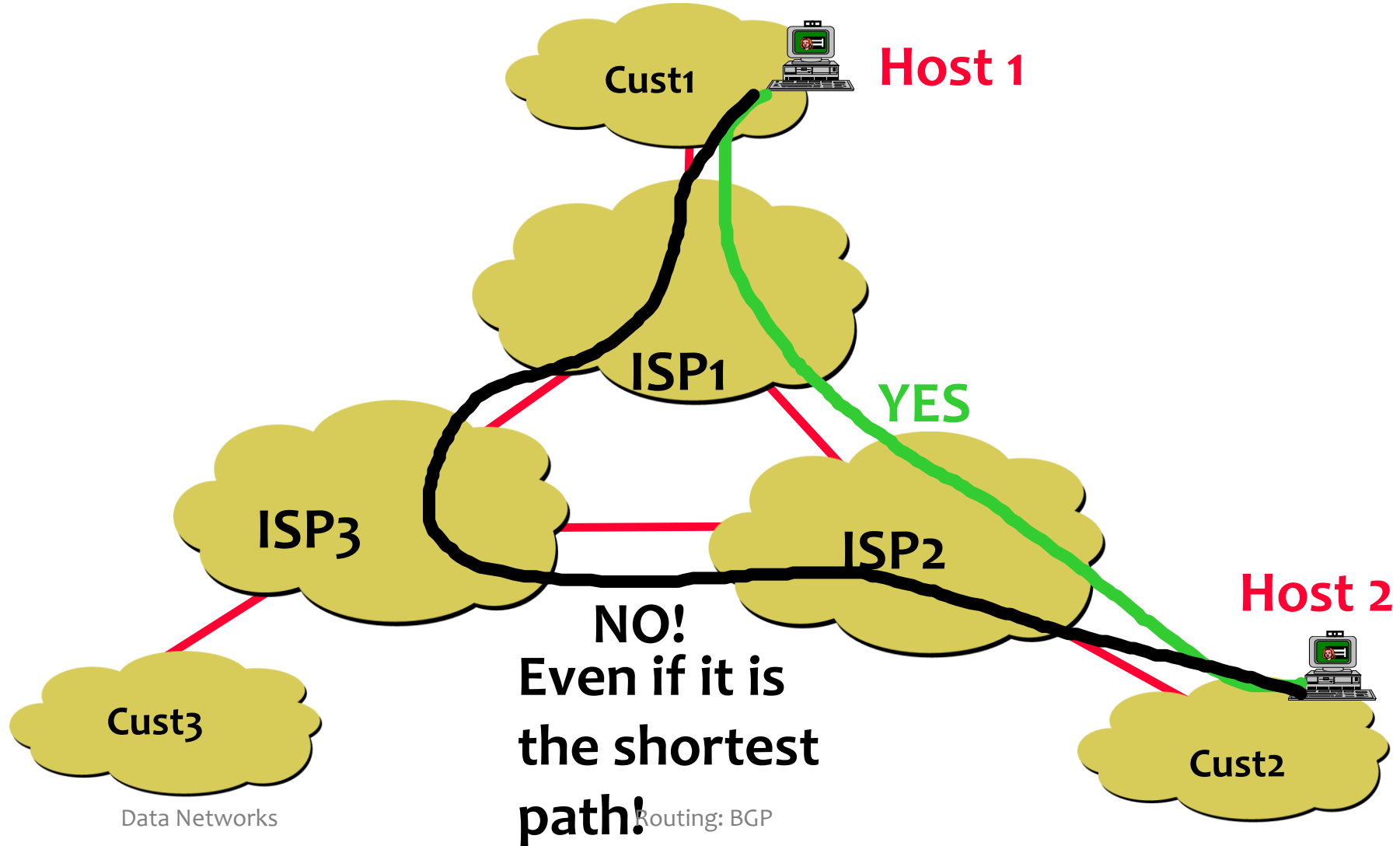
Routing policy: Examples



- **Honor business relationships**
(e.g., customers get full-table; peers only customer prefixes)
(e.g., prefer customer routes over peer routes over upstream routes)
- Allow customers a choice of route
(e.g., on customer request do not export prefix to AS x, etc.)
- Enable customer traffic engineering
(e.g., prepend x times to all peers or to specified AS)
- Enable DDoS defense for customers
(e.g., blackholing by rewriting the next hop)
- ...



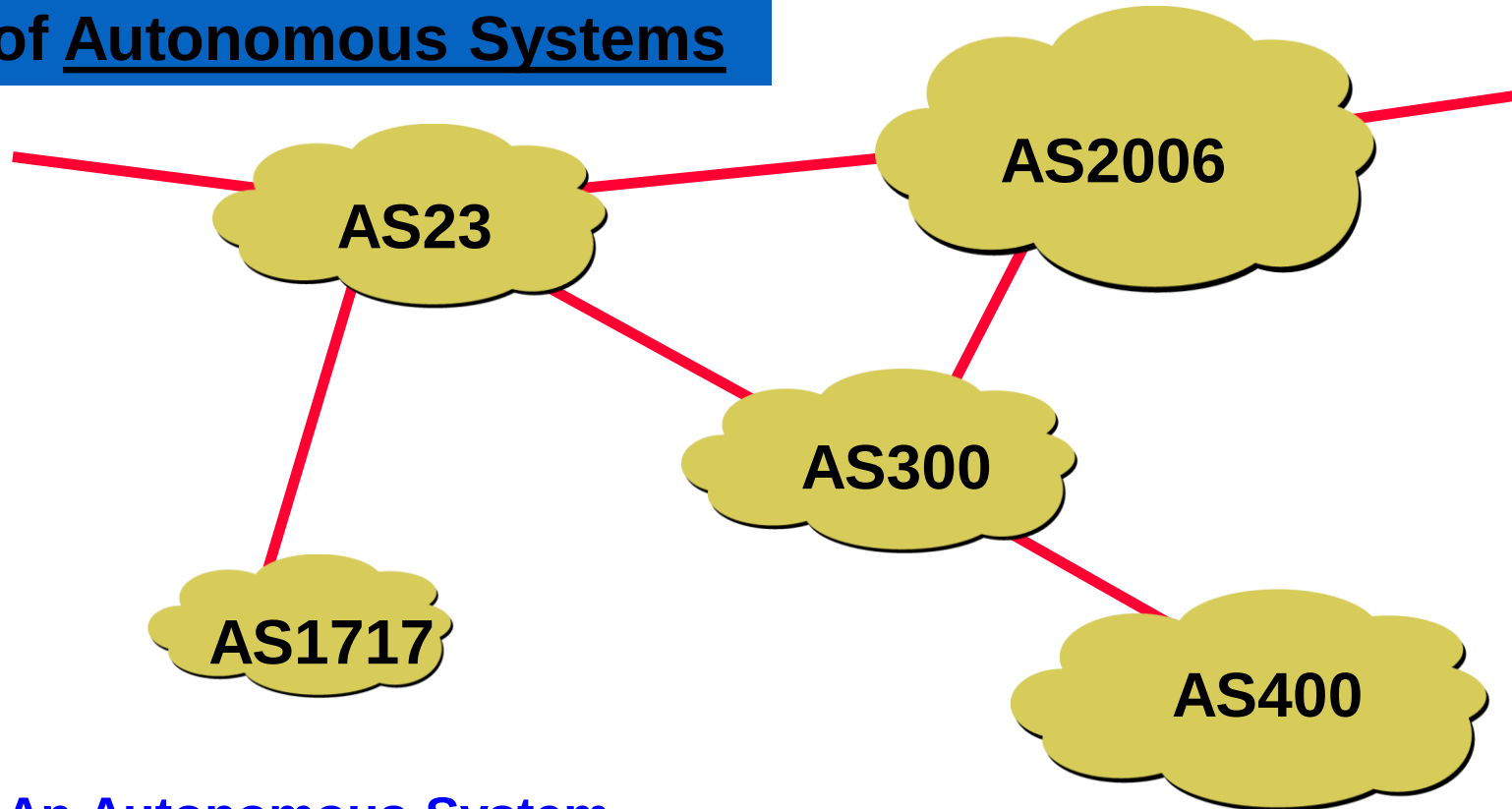
Why policy should win over distance metrics



Current Internet architecture



Arbitrary Internetwork of Autonomous Systems



**An Autonomous System
is a unified administrative
domain with a consistent
routing policy**

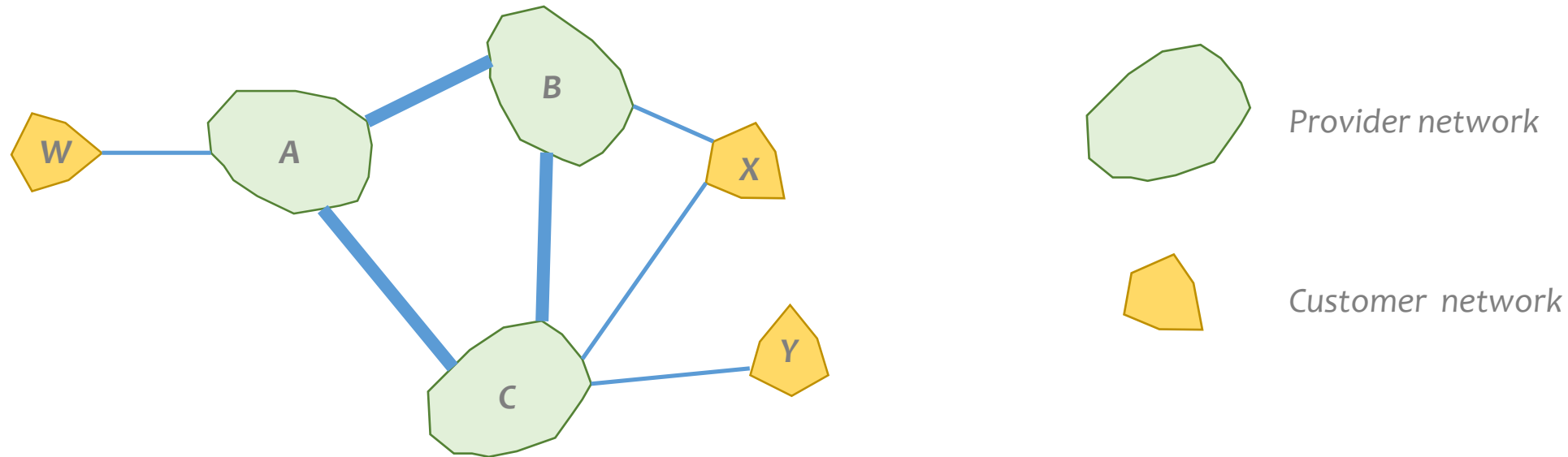
Data Networks

**A few years ago about 7000 AS
numbers are assigned,
about 4200 in use**

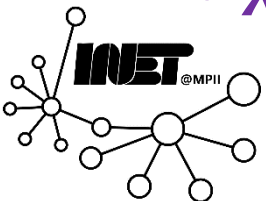
Routing: BGP



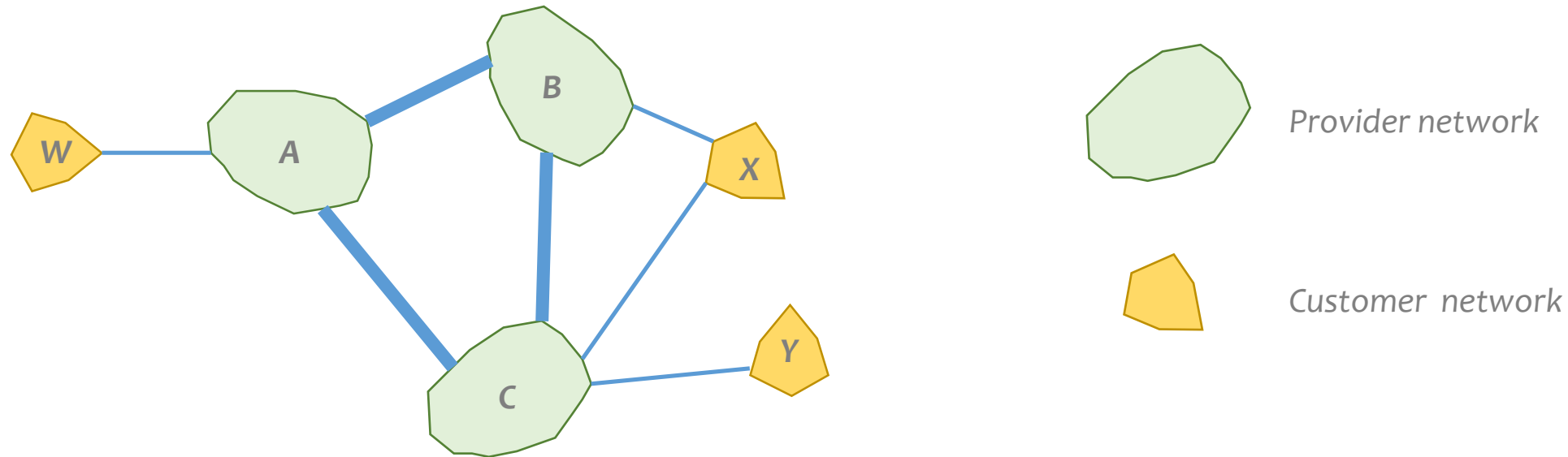
BGP routing policy



- **A**, **B**, and **C** are **provider** networks, and are peers
- **X**, **W**, **Y** are **customer** (of provider networks)
- **X** is **dual-homed** (i.e., attached to two networks)
- **X** does not want to route from **B** via **X** to **C**
 - ... so **X will not** advertise to **B** a route to **C**

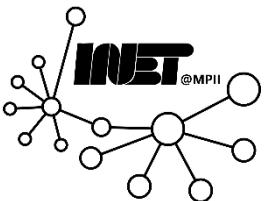
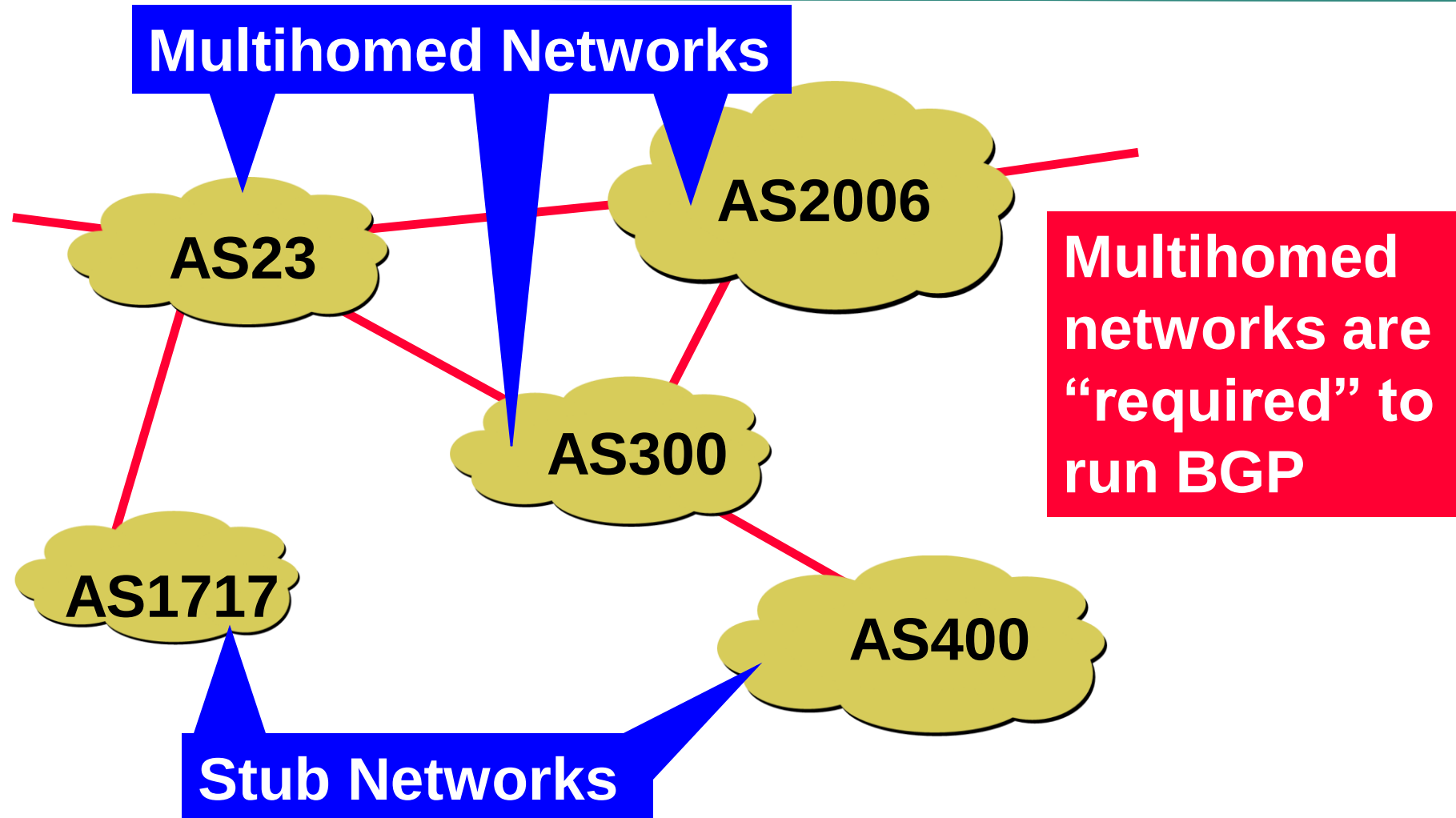


BGP routing policy

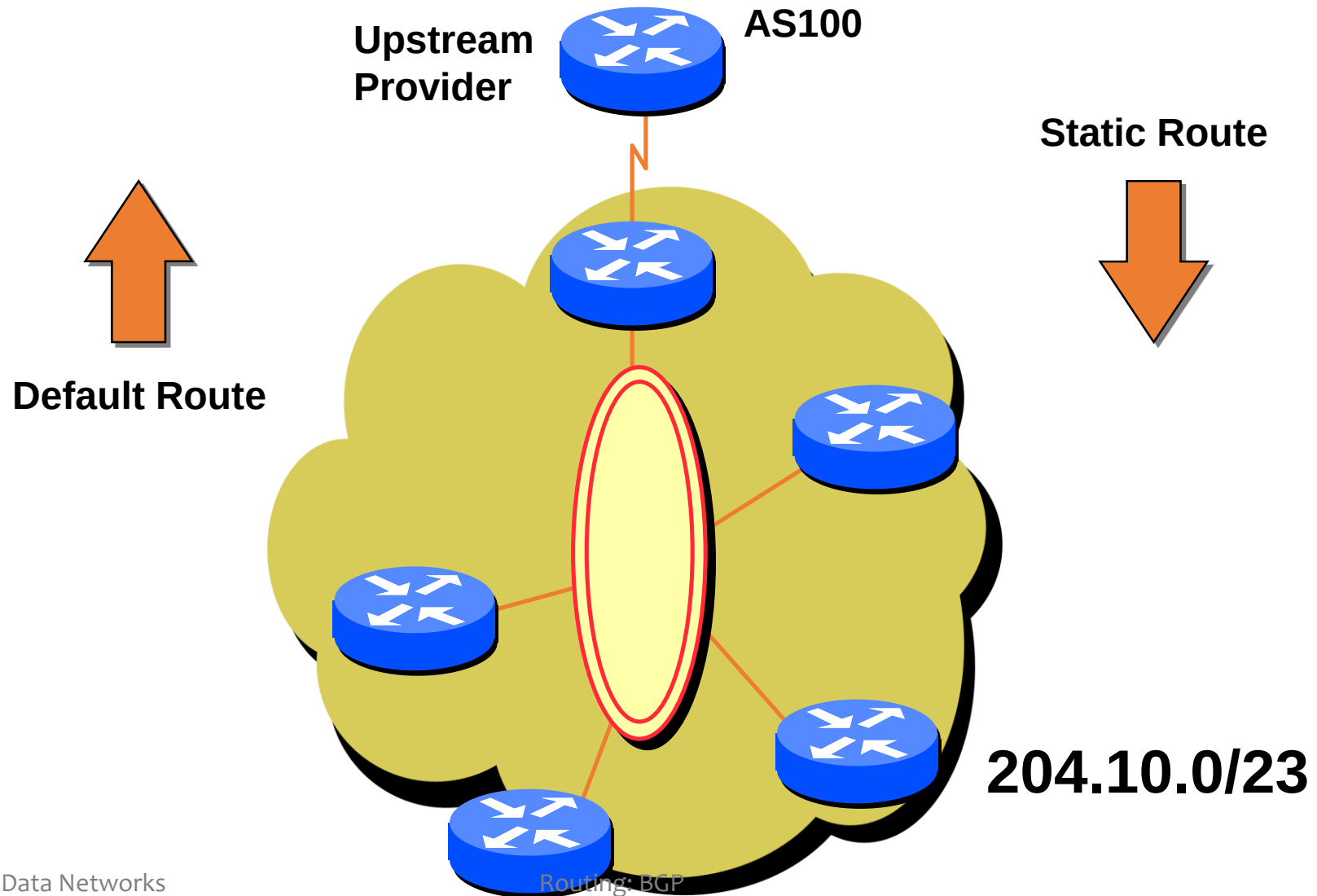


- **A** advertises to **B** the path **A-W**; **B** advertises to **X** the path **B-A-W**
- *Should B advertise to C the path BAW?*
 - **No way!** **B** gets no “revenue” for routing **C-B-A-W** since neither **W** nor **C** are **B**’s customers
 - **B** wants to **force** **C** to route to **W** via **A**
 - **B** wants to route **only** to and from its customers!

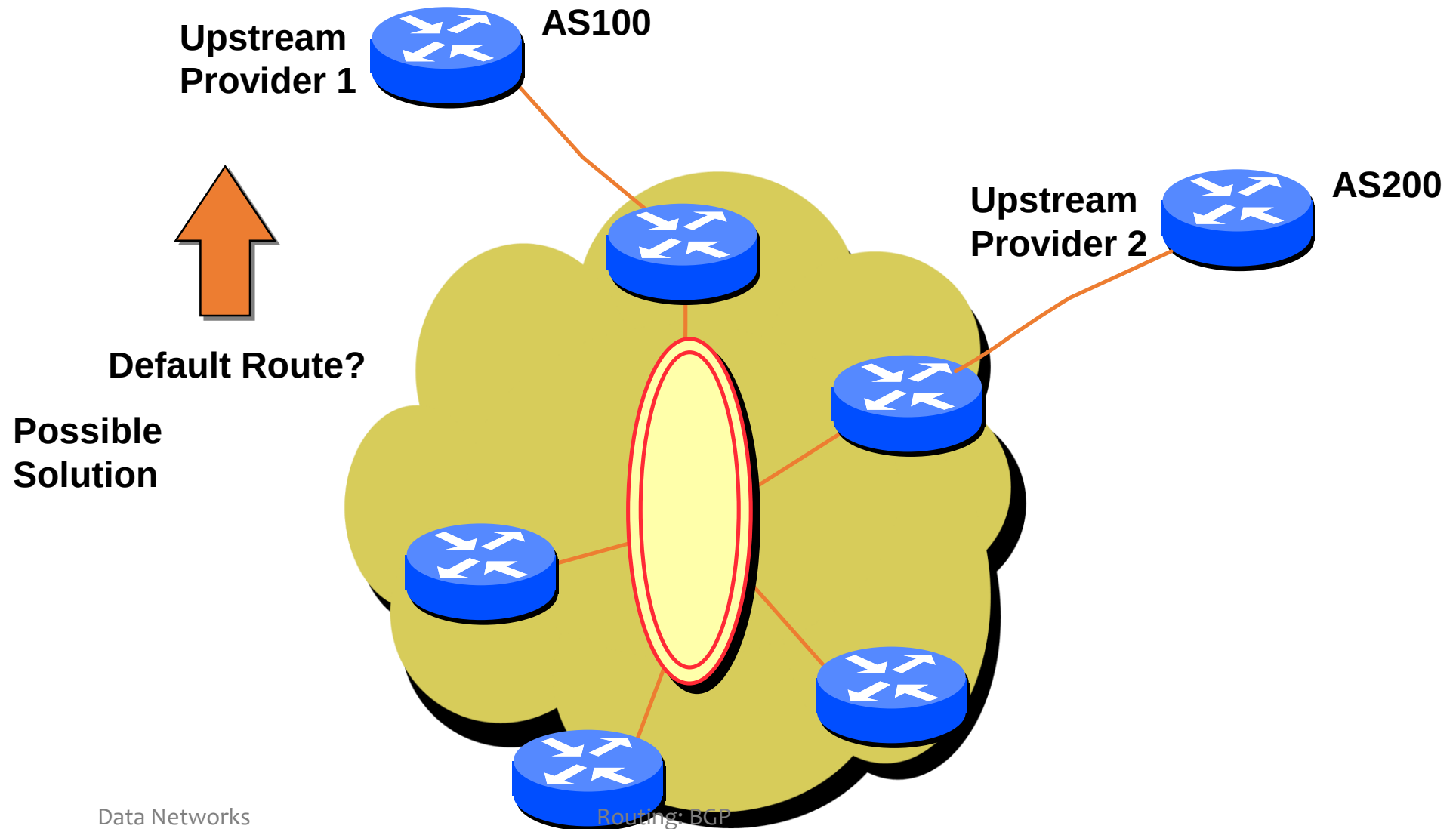
Stub vs. Multihomed networks



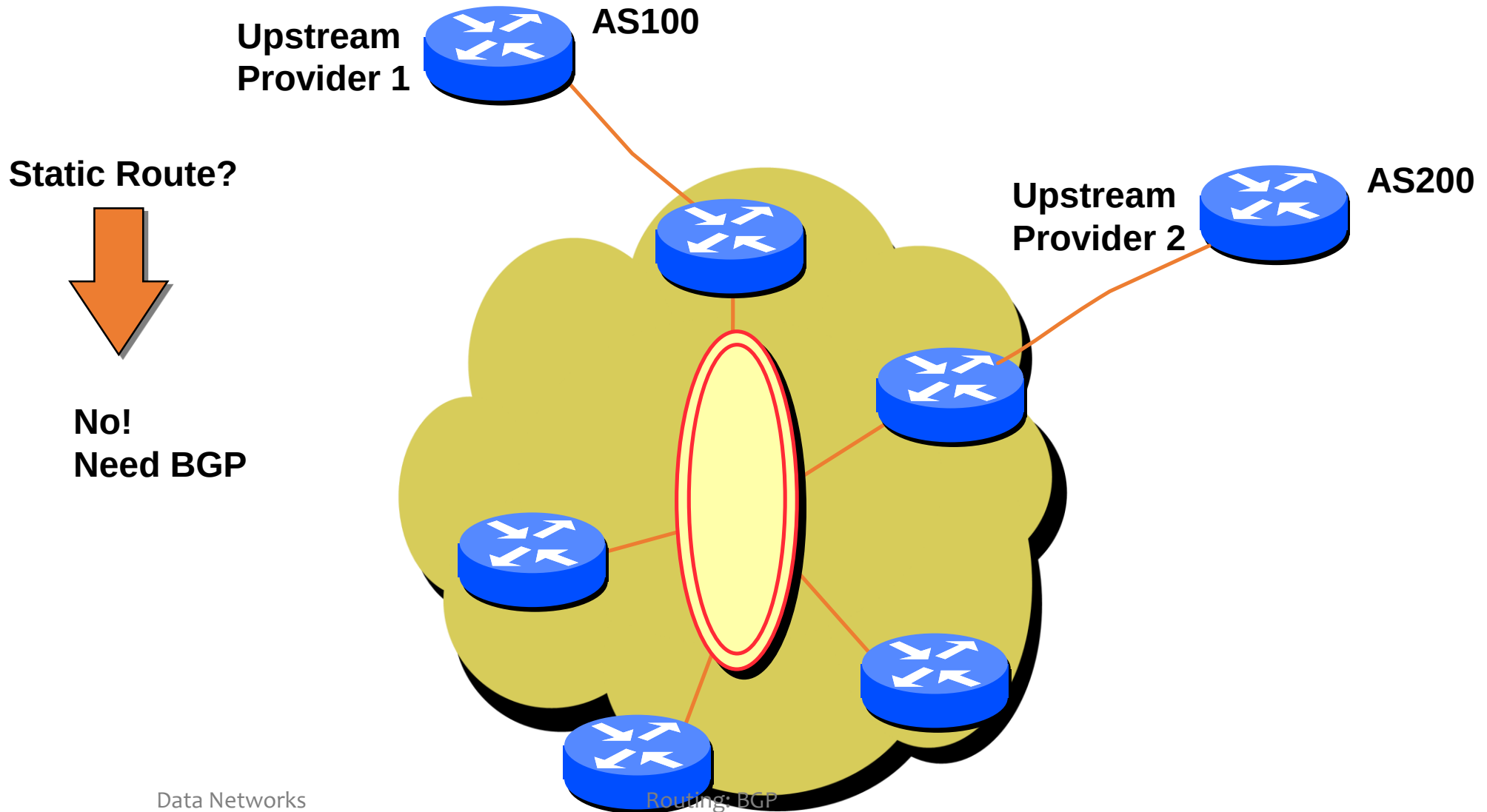
Routing at Stub ASs



Routing at Stub ASs – Multiple providers



Routing at Stub ASs – Multiple providers

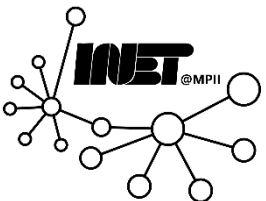
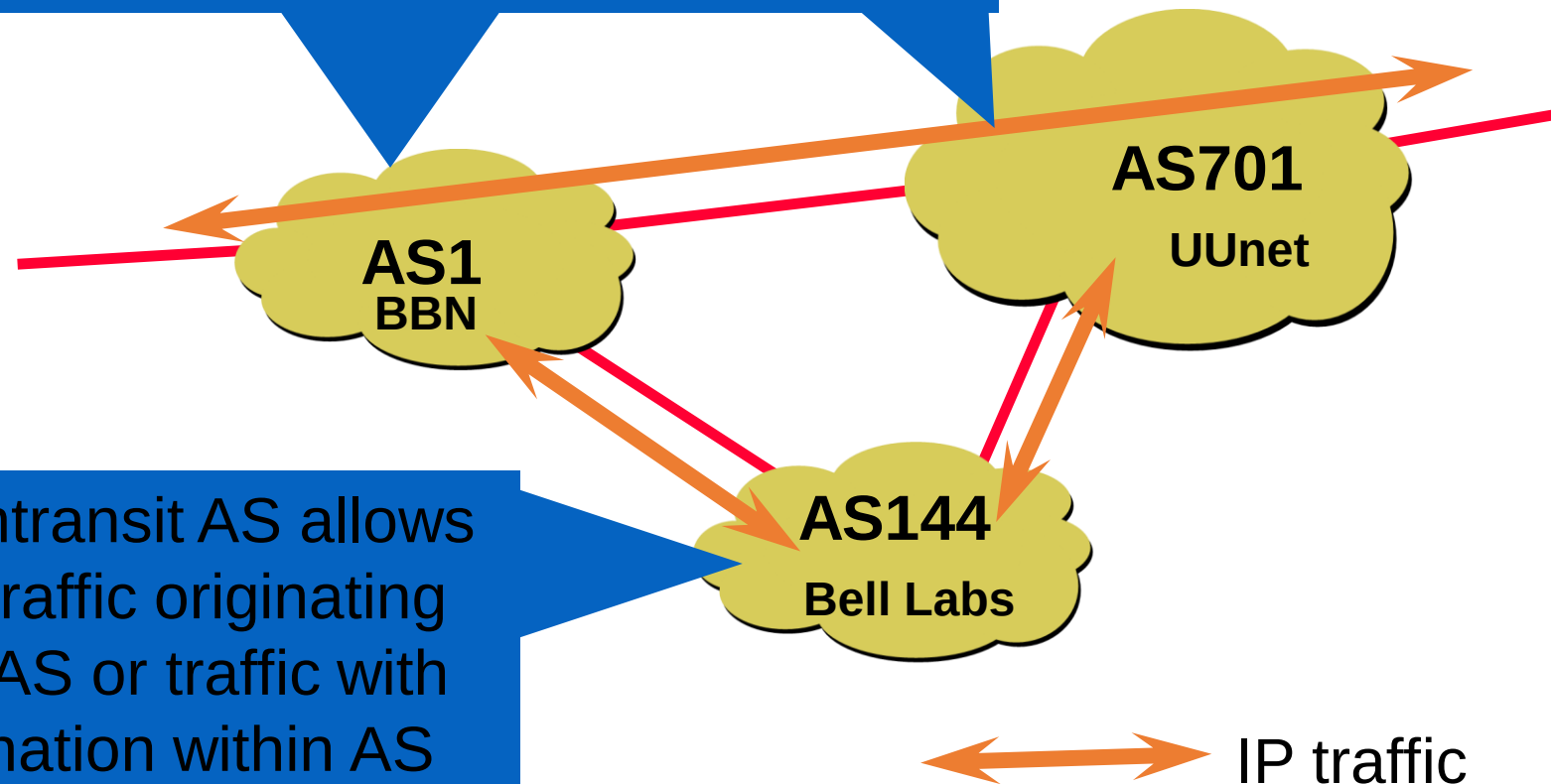


Policy: Transit vs. Nontransit

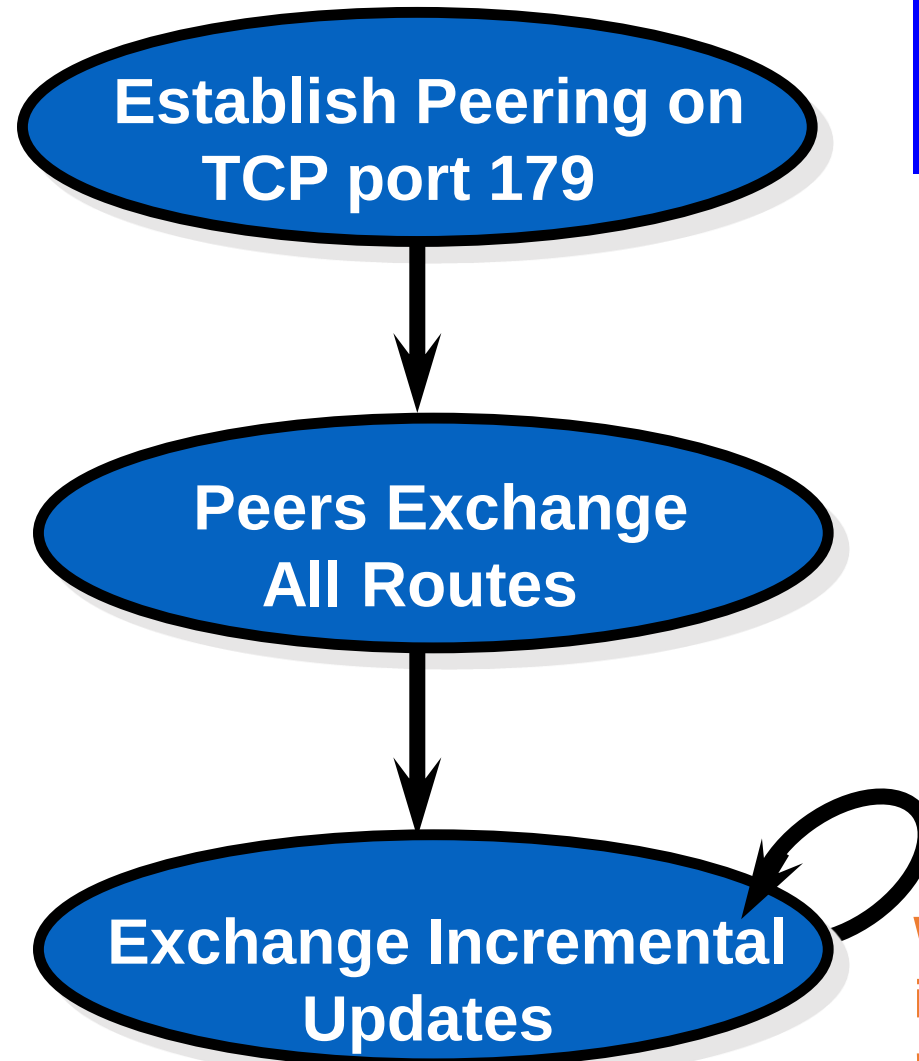


A transit AS allows traffic with neither source nor destination within AS to flow across the network

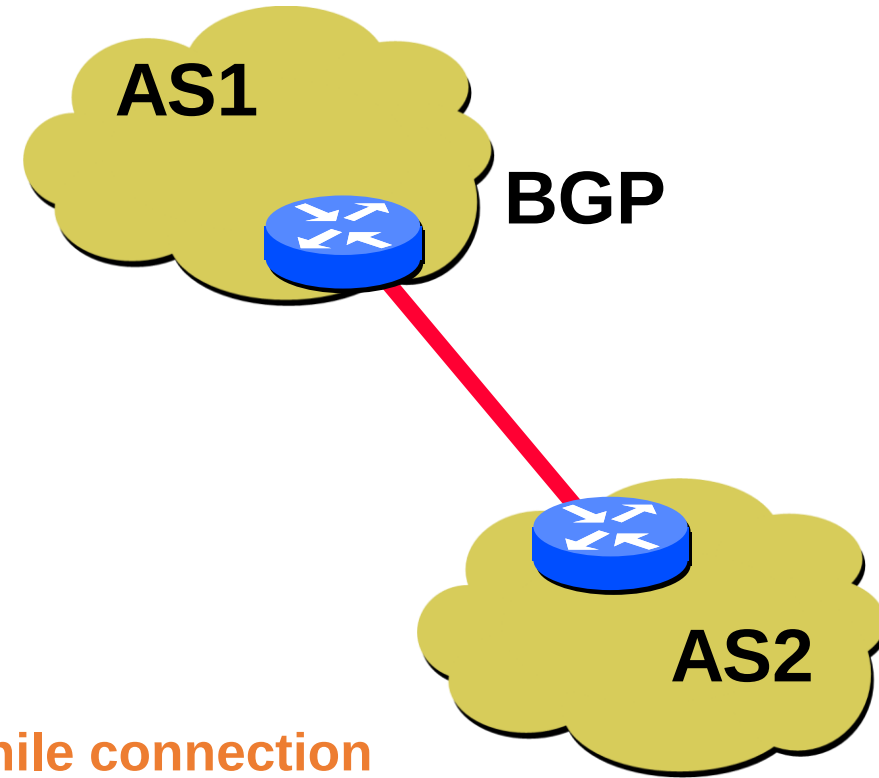
A nontransit AS allows only traffic originating from AS or traffic with destination within AS



BGP operations simplified



**BGP Route =
network prefix + attributes**



**While connection
is ALIVE exchange
route UPDATE messages**

BGP messages



Peers exchange **BGP messages** using **TCP**

- **OPEN:**
 - Opens TCP connection to peer
 - Authenticates sender
- **UPDATE:**
 - Advertises new routes (or withdraws old)
- **KEEPALIVE:**
 - Keeps conn alive in absence of UPDATES, ACKs OPEN request
- **NOTIFICATION:**
 - Reports errors in previous message; closes a connection

Process:

- **Initialization:** Open $\Rightarrow$ Updates for all routes
- **Ongoing:** Updates for changed routes

